

Binomial Distribution.

Using the binomial distribution
find the probability of

- (i) 3 successes in 8 trials when $p = 0.4$

$$n = 8$$

$$p = 0.4$$

$$q = 1 - 0.4$$

$$= 0.6$$

$$x = 3$$

$$P(X=3) = {}^n C_x p^x q^{n-x}$$

$$= {}^8 C_3 p^3 q^5$$

$$= 56 (0.4)^3 (0.6)^5$$

$$= 0.2786$$

- (ii) 2 failures in 6 trials when $p = 0.6$

$$p = 0.6$$

$$q = 0.4$$

$$n = 6$$

$$X = \text{failure} = 2$$

$$X = 6 - 2 = 4 \text{ success}$$

$$P(X=4) = {}^6 C_4 (0.6)^4 (0.4)^2$$

$$= 15 \times 0.1296 \times 0.16$$

$$= 0.311$$

(iii) 2 or fewer success in 9 trials
when $p = 0.4$

$$\begin{aligned} n &= 9 \\ p &= 0.4 \\ q &= 0.6 \end{aligned}$$

$$\begin{aligned} P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= [{}^9C_0 p^0 q^9 + {}^9C_1 p^1 q^8 + {}^9C_2 p^2 q^7] \\ &= [0.0101 + 0.0605 + 0.1613] \\ &= 0.2319 \end{aligned}$$

Q#02 If $n=15$ and $p=q$ then find

(i) $P(X=2)$

$$\begin{aligned} p &= q \\ p &= 1-p \\ 2p &= 1 \\ p &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} p &= 0.5 \\ \text{As } p &= q \text{ so } q = 0.5 \\ n &= 15 \end{aligned}$$

$$\begin{aligned} P(X=2) &= {}^{15}C_2 p^2 q^{13} \\ &= 105(0.5)^2(0.5)^{13} \\ &= 105(0.5)^{15} \end{aligned}$$

$$\begin{aligned} &= 105 \times 0.0000305 \\ &= 0.0032 \end{aligned}$$

(ii) $P(X \leq 2)$

$$\begin{aligned} P(X \leq 2) &= \\ 1 - [P(X=0) + P(X=1)] \end{aligned}$$

$$1 - [{}^{15}C_0 p^0 q^{15} + {}^{15}C_1 p^1 q^{14}]$$

$$1 - [0.1 \times 1(0.5)^{15} + 15(0.5)^1(0.5)^{14}]$$

$$1 - [0.0000305 + 15(0.5)^{15}]$$

$$\frac{0.0004515}{4515}$$

$$1 - [0.0000305 + 15 \times 0.0000305]$$

$$0.000488$$

$$= 0.9995$$

(iii) $P(\text{not more than } 3)$

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

From above values.

$$= [0.000488 + {}^{15}C_2 p^2 q^{13} + {}^{15}C_3 p^3 q^{12}]$$

$$= [0.000488 + 105 \times 0.0000305 + 455 \times 0.0000305]$$

$$= [0.000488 + 0.0032 + 0.01387]$$

$$= 0.0176$$

$$1 - \frac{11}{32} = \frac{10}{32}$$

Q#03 A coin is tossed 6 times what is the probability that head appears not more than 3 times.

$$n = 6$$

$$p = 0.5$$

$$q = 0.5$$

$$P(X \leq 3) = 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - [{}^6C_0 p^0 q^6 + {}^6C_1 p^1 q^5 + {}^6C_2 p^2 q^4]$$

$$= 1 - [1 \times 1 (0.5)^6 + 6 (0.5)^1 (0.5)^5 + 15 (0.5)^2 (0.5)^4]$$

$$= 1 - [1 (0.5)^6 + 6 (0.5)^6 + 15 (0.5)^6]$$

$$= 1 - (0.5)^6 [1 + 6 + 15]$$

$$= 1 - (0.5)^6 (22)$$

$$= 1 - 0.0156 \times 22$$

$$= 1 - 0.34375$$

$$= 0.65625$$

Q#04 Show for binomial random variable X with $n=3$ that $P(X=1 \text{ or } 2) = 3P(1-p)$

$$n=3$$

$$P(X=1 \text{ or } 2) = P(X=1) + P(X=2)$$

$$= {}^3C_1 p q^{3-1} + {}^3C_2 p^2 q^{3-2}$$

$$3p(1-p)^2 + 3p^2(1-p)$$

$$\frac{3p(1-p)[(1-p) + p]}{3p(1-p)}$$

so

$$P(X=1 \text{ or } 2) = 3P(1-p)$$

Q#05 Find the Binomial distribution when mean 12 & S.D = 3

$$np = 12 \text{ --- (i) } \sqrt{npq} = 3$$

$$npq = 9 \text{ --- (ii)}$$

Divide 1 & 2

$$\frac{npq}{np} = \frac{9}{12}$$

$$q = \frac{3}{4}$$

$$p = 1 - q$$

$$p = 1 - \frac{3}{4}$$

$$p = \frac{1}{4}$$

$$p = \frac{1}{4}$$

from (i)

$$np = 12$$

$$n\left(\frac{1}{4}\right) = 12$$

$$n = 12 \times 4$$

$$n = 48$$

$$\begin{aligned} B.O &= (q + p)^n \\ &= \left(\frac{3}{4} + \frac{1}{4}\right)^{48} \\ &= \left(\frac{4}{4}\right)^{48} \\ &= (1)^{48} \\ &= 1 \end{aligned}$$

Q#6 In a BD with $n=4$ what value can p have if $p(0) = p(1)$

$$n=4$$

$$P(X=0) = P(X=1)$$

$${}^4C_0 p^0 q^4 = {}^4C_1 p^1 q^3$$

$$(1-p)^4 = 4p(1-p)^3$$

Divide b/s by $(1-p)^3$

$$(1-p) = 4p$$

$$1 = 4p + p$$

$$1 = 5p$$

$$p = \frac{1}{5} = 0.2$$

Q#7 A random variable is binomially distributed with mean 12.38 and variance 8.64. Find n & p

$$np = 12.38 \quad \text{--- (i)}$$

$$npq = 8.64 \quad \text{--- (ii)}$$

from (i)

$$npq = 8.64$$

$$np(1-p) = 8.64$$

$$12.38(1-p) = 8.64$$

$$1-p = \frac{8.64}{12.38}$$

$$1-p = 0.6978$$

$$p = 1 - 0.6978$$

$$p = 0.3022$$

from (i)

$$np = 12.38$$

$$n(0.3022) = 12.38$$

$$n = \frac{12.38}{0.3022}$$

$$n = 41$$

Q#8 In a BD of order 7 $P(X=2) = P(X=3)$
Find $P(X=4)$ & $P(X \leq 2)$

$$n=7$$

$$P(X=2) = P(X=3)$$

$${}^7C_2 p^2 q^5 = {}^7C_3 p^3 q^4$$

$$21p^2(1-p)^5 = 35p^3(1-p)^4$$

Divide b/s by $p^2(1-p)^4$

$$21(1-p) = 35p$$

$$21 - 21p = 35p$$

$$21 = 35p + 21p$$

$$21 = 56p$$

$$p = \frac{21}{56} = \frac{3}{8} \quad q = \frac{5}{8}$$

Now calculate $P(x=4)$

$$\begin{aligned} {}^7C_4 p^4 q^3 &= 35 \left(\frac{3}{8}\right)^4 \left(\frac{5}{8}\right)^3 \\ &= \frac{125}{512} \approx 0.169 \end{aligned}$$

$$\begin{aligned} P(x \leq 2) &= 1 - [P(x=0) + P(x=1)] \\ &= 1 - [{}^7C_0 p^0 q^7 + {}^7C_1 p^1 q^6] \end{aligned}$$

$$= 1 - \left[\left(\frac{5}{8}\right)^7 + 7 \left(\frac{3}{8}\right) \left(\frac{5}{8}\right)^6 \right]$$

$$= 1 - [0.0373 + 0.156]$$

$$= 0.8067$$

Q#09 What is the mathematical expectation of the no of heads when 3 fair coins are tossed.

$$n = 3$$

$$\text{Head (H)} = \frac{1}{2}$$

$$\text{Tail (T)} = \frac{1}{2}$$

let x be the no of heads in 3 tosses.

Then the BD with $n=3, p=\frac{1}{2}$

$$\begin{aligned} E(X) &= n p \\ &= 3 \cdot \frac{1}{2} \end{aligned}$$

$$= \frac{3}{2}$$

$$= 1.5$$

Q#10 A die is rolled 5 times and a 4 or 5 is considered a success. Find the probability of

(i) no success

$$\begin{aligned} n &= 5 \\ p &= \frac{2}{6} = \frac{1}{3} \quad q = 1 - p = \frac{2}{3} \end{aligned}$$

$$P(x=0) = {}^5C_0 p^0 q^5$$

$$= \left(\frac{2}{3}\right)^5$$

$$= 0.1317$$

(ii) at least 3 successes.

$$P(x \geq 3)$$

$$\begin{aligned} &= 1 - [P(x=0) + P(x=1) + P(x=2)] \\ &= 1 - \left[\left(\frac{2}{3}\right)^5 + 5 \left(\frac{2}{3}\right)^4 + 10 \left(\frac{2}{3}\right)^3 \right] \\ &= 1 - \left(\frac{2}{3}\right)^5 [1 + 5 + 10] \\ &= 1 - \left(\frac{2}{3}\right)^5 (16) \end{aligned}$$

(ii) at least 4 but not more than 3
3 successes

$$P(X \geq 3) = 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - [{}^5C_0 p^0 q^5 + {}^5C_1 p^1 q^4 + {}^5C_2 p^2 q^3]$$

$$= 1 - [1 \times 1 \left(\frac{2}{3}\right)^5 + 5 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^4 + 10 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^3]$$

$$= 1 - \left[\frac{32}{243} + \frac{80}{243} + \frac{80}{243} \right]$$

$$= 1 - \frac{192}{243}$$

$$= 0.2099$$

Q#11 Find the prob of obtaining exactly three services within the 4 services if $p = 0.60$

$$n = 4$$

$$x = 3$$

$$p = 0.60$$

$$q = 0.40$$

$$P(X=3) = {}^4C_3 p^3 q^1$$

$$= 4 (0.60)^3 (0.40)$$

$$= 4 (0.216) (0.40)$$

$$= 4 \cdot 0.0864$$

$$= 0.3456$$

Q#12 Suppose that warranty records show the prob that the new car needs warranty repair in the first 6 months is 0.8. If a sample of 5 cars is selected. What is the prob that

(i) None needs a warranty repair

$$n = 5$$

$$p = 0.8$$

$$q = 0.2$$

$$P(X=0) = {}^5C_0 p^0 q^5$$

$$= (0.2)^5$$

$$= 0.00032$$

(ii) At least 3 needs a warranty repair

$$P(X \geq 3) = P(3) + P(4) + P(5)$$

$$P(X=3) = {}^5C_3 p^3 q^2 + {}^5C_4 p^4 q^1 + {}^5C_5 p^5 q^0$$

$$= 10 \cdot 0.512 \cdot 0.04 + 5 \cdot 0.4096 \cdot 0.2 + 1 \cdot 0.32768$$

$$= 0.2048 + 0.4096 + 0.32768$$

$$= 0.94208$$

(iii) More than one needs a warranty repair

$$\begin{aligned}
 P(x > 1) &= 1 - [P(0) + P(1)] \\
 &= 1 - [0.00032 + {}^5C_1 p^1 q^4] \\
 &= 1 - [0.00032 + 5(0.8)(0.2)^4] \\
 &= 1 - [0.00032 + 0.0064] \\
 &= 0.99328
 \end{aligned}$$

(iv) Mean & variance

$$\begin{aligned}
 \text{Mean} = np &= 5(0.8) \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 \text{Variance} &= npq \\
 &= 5(0.8)(0.2) \\
 &= 0.8
 \end{aligned}$$

Q#3 Ten families having 3 children each are selected at random from the 1000 families. Find.

(i) Prob that the selected ten are girls if we take each person per home assuming of 1 girl and 2 boys each family

$$n = 10$$

Prob of picking the girl = $\frac{1}{3}$

prob of boy = $\frac{2}{3}$

$$P(\text{all ten are girls}) = \left(\frac{1}{3}\right)^{10} = 0.0000169$$

(ii) Mean & variance

$$n = 10$$

$$p = \frac{1}{3} \quad q = \frac{2}{3}$$

$$\text{Mean} = np = \frac{10 \times 1}{3} = \frac{10}{3} = 3.33$$

$$\begin{aligned}
 \text{Variance} = npq &= \frac{10 \times 1 \times 2}{3 \times 3} \\
 &= 22.2
 \end{aligned}$$

Q#4 A class consists of 50 students. If on a particular day 5 students are selected what is the prob that the 5 will be 3 boys if there are 30 boys in the class?

$$\text{Total students} = 50$$

$$\text{Boys} = 30$$

$$\text{Girls} = 20$$

$n = 5$
prob of selecting a boy = $\frac{30}{50}$

$$q = 0.4 \quad p = 0.6$$

$$P(X=3) = {}^5C_3 p^3 q^2$$

$$= 10 (0.6)^3 (0.4)^2$$

$$= 10 (0.216) (0.16)$$

$$= 0.3456$$

Q#15 The prob that a person can hit a bull on a target is $\frac{1}{3}$ is he certain to score a bull if he takes 3 shots? Also find the prob of his scoring at least 1 bull in 3 shots?

$$p = \frac{1}{3} \quad q = \frac{2}{3} \quad n = 3$$

Is he certain to hit the bull in 3 shots

No, certain would mean 100% prob but he can also miss 3 shots so,

$$\begin{aligned} P(X=0) &= {}^3C_0 p^0 q^3 \\ &= 1 \cdot 1 \left(\frac{2}{3}\right)^3 \\ &= \frac{8}{27} \end{aligned}$$

Prob of scoring at least 1 in 3

$$\begin{aligned} P(X \geq 1) &= 1 - [P(X=0)] \\ &= 1 - \frac{8}{27} = \frac{19}{27} \end{aligned}$$

Q#16 Aslam tossed a coin 10 times find the prob of getting exactly 7 heads or no heads?

$$n = 10$$
$$p = \frac{1}{2}, \quad q = \frac{1}{2}$$

$$\begin{aligned} P(X=7) &= {}^{10}C_7 p^7 q^3 \\ &= 120 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 \\ &= 120 \left(\frac{1}{2}\right)^{10} \\ &= 120 \times \frac{1}{1024} \\ &= 0.1172 \end{aligned}$$

No heads

$$\begin{aligned}P(X=0) &= {}^{10}C_0 p^0 q^{10} \\&= 1 \times 1 \left(\frac{1}{2}\right)^{10} \\&= \frac{1}{1024}\end{aligned}$$

Q#17 A die is rolled 5 times. We get a six. What is the chance that out of 5 there are 4 sixes or no sixes

$$\begin{aligned}n &= 5 \\p &= \text{No of success is getting a six} = \frac{1}{6} \\q &= \frac{5}{6}\end{aligned}$$

$$\begin{aligned}P(X=4) &= {}^5C_4 p^4 q^1 \\&= 5 \cdot \left(\frac{1}{6}\right)^4 \frac{5}{6} \\&= \frac{25}{7776} = 0.00321\end{aligned}$$

$$\begin{aligned}P(X=0) &= {}^5C_0 p^0 q^5 \\&= 1 \times 1 \left(\frac{5}{6}\right)^5 \\&= 0.4051\end{aligned}$$

Q#18 An insurance salesperson sells policies to 5 men all of identical age and in good health. Acc to the actuarial table the prob that a man of his particular age will be alive 30 years hence is $\frac{2}{3}$. Find the prob that in 30 years

(i) All 5 men will be alive

$$\begin{aligned}n &= 5 \\p &= \frac{2}{3} \\q &= \frac{1}{3} \\P(X=5) &= {}^5C_5 p^5 q^0 \\&= 1 \cdot \left(\frac{2}{3}\right)^5 \\&= \frac{32}{243} \approx 0.1317\end{aligned}$$

(ii) At least 1 man will be alive

$$\begin{aligned}P(X \geq 1) &= 1 - [P(X=0)] \\&= 1 - [{}^5C_0 p^0 q^5] \\&= 1 - \left[1 \times 1 \left(\frac{1}{3}\right)^5\right] \\&= 1 - \left[\frac{1}{243}\right] \\&= \frac{242}{243} \approx 0.996\end{aligned}$$

iii Not more than 3 will be alive

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= \left[\frac{1}{243} + \frac{10}{243} + \frac{40}{243} + \frac{80}{243} \right]$$

$$= \frac{131}{243}$$

$$= 0.5391$$

Q#19 The incidence of occupational disease in an industry is such that the workman has a 30% chance of suffering from it. What is the prob that out of 8 workman

(i) Not more than 2 will catch the disease

$$n = 8$$

$$p = 0.3$$

$$q = 0.7$$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= {}^8C_0 p^0 q^8 + {}^8C_1 p^1 q^7 + {}^8C_2 p^2 q^6$$

$$= (0.7)^8 + 8(0.3)(0.7)^7 + 28(0.3)^2(0.7)^6$$

$$= 0.0576 + 0.1978 + 0.2964$$

$$= 0.5518$$

(ii) 4 or more will catch the disease

$$P(X \geq 4) = 1 - P(X \leq 3)$$

We already have $P(X \leq 3) = 0.5518$

$$P(X=3) = {}^8C_3 p^3 q^5$$

$$= 0.2543$$

$$P(X \leq 3) = 0.5518 + 0.2543$$

$$= 0.8061$$

$$P(X \geq 4) = 1 - 0.8061$$

$$= 0.1939$$

Q#20 A certain event is believed to follow the BD. In 1204 samples of 5, the result was observed once 405 times and twice 270 times Find p & q

$$n = 5$$

$$\text{Total samples} = 1204$$

$$\text{Event happened 1 : 405 samples } X=1$$

$$\text{Event happened 2 : 270 samples } X=2$$

$$P(X=1) = {}^5C_1 p^1 q^4$$

$$= 5pq^4$$

$$P(X=2) = {}^5C_2 p^2 q^3$$

$$= 10p^2 q^3$$

Since there are 1204 samples.

$$P(X=1) \approx \frac{405}{1204} = 0.3364$$

$$P(X=2) \approx \frac{270}{1204} = 0.2242$$

$$\frac{P(X=2)}{P(X=1)} = \frac{10p^2q^3}{5pq^4}$$

$$\frac{0.2242}{0.3364} = \frac{2p}{q}$$

$$\frac{2p}{q} = 0.6667$$

$$\frac{2p}{1-p} = 0.6667$$

$$2p = 0.6667(1-p)$$

$$2p = 0.6667 - 0.6667p$$

$$2.6667p = 0.6667$$

$$p = \frac{0.6667}{2.6667}$$

$$p = 0.25$$

$$q = 1 - p$$

$$= 1 - 0.25$$

$$q = 0.75$$

Q#21 The prob distribution of a discrete random variable X is

$$f(x) = \binom{3}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}$$

$X = 0, 1, 2, 3$ Find $E(X)$ and $E(X^2)$

$$n = 3$$

$$p = \frac{1}{4}$$

$$q = \frac{3}{4}$$

$$E(X) = np = 3 \cdot \frac{1}{4} = \frac{3}{4}$$

$$E(X^2) = \text{Var}(X) + [E(X)]^2$$

$$\text{Var}(X) = npq = 3 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{9}{16}$$

$$\text{so } E(X^2) = \frac{9}{16} + \left(\frac{3}{4}\right)^2$$

$$= \frac{9}{16} + \frac{9}{16}$$

$$= \frac{18}{16}$$

$$= \frac{9}{8}$$

Q#22 A student's prob in each subject is $\frac{3}{4}$ (75%). He appears in 5 subjects in annual examination. What is the prob that

(i) he fails in 3 subjects

$$\begin{aligned}
 q &= \frac{3}{4} & p &= \frac{1}{4} \\
 n &= 5 \\
 P(X=3) &= {}^5C_3 p^3 q^2 \\
 &= 10 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 \\
 &= 10 \frac{1}{64} \frac{9}{16} \\
 &= \frac{90}{1024} \\
 &= \frac{45}{512} \approx 0.0879.
 \end{aligned}$$

(ii) He passes at least four subjects

$$\begin{aligned}
 p &= \frac{3}{4} & q &= \frac{1}{4} \\
 n &= 5 \\
 P(X \geq 4) &= P(X=4) + P(X=5) \\
 &= {}^5C_4 p^4 q^1 + {}^5C_5 p^5 q^0 \\
 &= 5 \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)^1 + \left(\frac{3}{4}\right)^5 \\
 &= \frac{905}{1024} + \frac{243}{1024} = \frac{81}{128} \\
 &\approx 0.6328
 \end{aligned}$$

(iii) he fails in at most two subjects

$$\begin{aligned}
 p &= \frac{3}{4} & q &= \frac{1}{4} & n &= 5 \\
 P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\
 &= \frac{1}{1024} + \frac{5}{1024} + \frac{15}{1024} \\
 &= \frac{21}{1024} \\
 &= \frac{808}{1024} = \frac{101}{128} \approx 0.78125
 \end{aligned}$$

(iv) successful in more than 1 subject

$$\begin{aligned}
 p &= \frac{3}{4} & q &= \frac{1}{4} & n &= 5 \\
 P(X > 1) &= 1 - [P(X=0) + P(X=1)] \\
 &= 1 - \left[\frac{1}{1024} + \frac{5}{1024} \right] \\
 &= 1 - \frac{6}{1024} \\
 &\approx 0.9844
 \end{aligned}$$

Q#23 A committee of size 5 is to be selected at random from 3 women & 5 men. Find the expected no of women on the committee.

3 women
5 men
Total = 8

$$\begin{aligned}
 &{}^8C_2 \\
 &10 \left(\frac{3}{8}\right)^2 \left(\frac{1}{4}\right) \\
 &\frac{10 \cdot 9}{16 \cdot 64} = \frac{90}{1024}
 \end{aligned}$$

²⁴
Q#20 A recent study found that 4 out of 9 houses are underinsured. If 5 houses are selected from the 9 houses find the prob that exactly 2 are underinsured.

$$\begin{aligned}
 n &= 5 \\
 p &= \frac{4}{9}, \quad q = \frac{5}{9} \\
 P(X=2) &= {}^5C_2 p^2 q^3 \\
 &= 10 \left(\frac{4}{9} \right)^2 \left(\frac{5}{9} \right)^3 \\
 &= \frac{10 \cdot 16 \cdot 125}{81 \cdot 729} \\
 &= \frac{20000}{59049} \\
 &= 0.3387
 \end{aligned}$$

²⁵
Q#21 A machine produces parts which are graded A or B acc to finish and the prop of A to B is well established at 2:1 calculate the probabilities of obtaining 0, 1, 2, 3, 4, 5 parts of finish A in a sample of 5.

$$\begin{aligned}
 \text{Propotion of A to B} &= 2:1 \\
 \text{Prob of A (sucess)} &= \frac{2}{3} \\
 \text{Prob of B (failure)} &= \frac{1}{3}
 \end{aligned}$$

Sample size $n = 5$

We are asked to calculate the BP for getting 0 to 5 parts of type A in a sample of 5

$$n = 5, r = 0, 1, 2, 3, 4, 5 \quad p = \frac{2}{3}, q = \frac{1}{3}$$

$$P(0) = {}^5C_0 p^0 q^5 = 0.0041$$

$$P(1) = {}^5C_1 p^1 q^4 = 0.0412$$

$$P(2) = {}^5C_2 p^2 q^3 = 0.1646$$

$$P(3) = {}^5C_3 p^3 q^2 = 0.3292$$

$$P(4) = {}^5C_4 p^4 q^1 = 0.3292$$

$$P(5) = {}^5C_5 p^5 q^0 = 0.1317$$

Q#26 Find BD with mean 2 & SD 1

$$np = 2 \quad \text{--- (i)}$$

$$npq = 1 \quad \text{--- (ii)}$$

$$\frac{npq}{np} = \frac{1}{2}$$

$$q = \frac{1}{2}$$

$$p = 1 - q = 1 - \frac{1}{2} = \frac{1}{2}$$

$$p = \frac{1}{2}$$

$$np = 2 = \frac{n \cdot 1}{2} = 2 = n = 4$$

~~n = 2~~

$$BD = (q + p)^n$$

$$= \left(\frac{1}{2} + \frac{1}{2} \right)^4$$

$$= (1)^4$$

$$= 1$$

Q27 A & B plays a game in which A's prob of winning is $\frac{2}{3}$. In a series of B games, what is the prob that A will win.

(i) exactly 4 games.

Q #28 If on the average rain falls of 10 days in every 30. find the prob that the first three days will be fine and the remaining wet

$$P(\text{rain}) = \frac{10}{30} = \frac{1}{3} \quad P(\text{no rain}) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$n = 7 \quad r = 3$$

$$P(x=3) = {}^nC_3 p^3 q^4$$

$$= {}^{35}C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^4$$

$$= \frac{35 \cdot 8}{27 \cdot 81}$$

$$= \frac{280}{2187}$$

$$= 0.128$$

if exactly three days are fine then
p of success = $\frac{2}{3}$

Again

$$P(\text{rain}) = \frac{1}{3}$$

$$P(\text{no rain}) = \frac{2}{3}$$

$$P = \left(\frac{2}{3}\right)^3 \cdot \left(\frac{1}{3}\right)^4$$

$$= \frac{8}{27} \cdot \frac{1}{81}$$

$$= \frac{8}{2187}$$

$$= 0.00366$$

(ii) Rain fall on just 3 days of week

Total possible combinations of 3 rainy days out of 7

$${}^7C_3 = 35$$

Favourable outcomes: only one specific combination when rain falls on day 1, 2 & 3.

$$P(\text{rain on day 1, 2, 3}) = \frac{1}{35} = 0.0286$$

Q#29 Experience shows that 80% of the people entering a store make a purchase. Find the prob that out of 6 people entering the store.

(i) no person will make a purchase

Prob of person making a purchase = $p = 0.8$

$$q = 0.2$$

$$n = 6$$

$$P(X=0) = {}^6C_0 p^0 q^6$$

$$= (0.2)^6$$

$$= 0.000064$$

(ii) exactly 4 will make a purchase

$$P(X=4) = {}^6C_4 p^4 q^2$$

$$= 15(0.8)^4(0.2)^2$$

$$= 15(0.4096)(0.04)$$

$$= 0.24576$$

(iii) at least 4 will make a purchase

$$P(X \geq 4) = P(X=4) + P(X=5) + P(X=6)$$

$$= {}^6C_4 p^4 q^2 + {}^6C_5 p^5 q^1 + {}^6C_6 p^6 q^0$$

$$= 0.24576 + 0.39322 + 0.26214$$

$$= 0.9011$$

(iv) at most 4 will make a purchase

$$P(X \leq 4) = 1 - [P(X=5) + P(X=6)]$$

$$= 1 - P(X=5) - P(X=6)$$

$$= 1 - {}^6C_5 p^5 q^1 - {}^6C_6 p^6 q^0$$

$$= 1 - 0.393 - 0.262$$

$$= 0.3447$$

Q#30

The Pakistan Postal service reports 95 percent of first class mail within the same city is delivered within two days of the time of mailing. Six letters are randomly sent to different locations.

(i) What is the probability that all six arrive within two days?

$$p = 0.95$$

$$q = 0.05$$

$$n = 6$$

$$\begin{aligned} P(X=6) &= {}^6C_6 p^6 q^0 \\ &= 1(0.95)^6 \\ &= 0.7351 \end{aligned}$$

(ii) Prob that exactly 5 arrive within 2 days?

$$\begin{aligned} P(X=5) &= {}^6C_5 p^5 q^1 \\ &= 6(0.95)^5(0.05)^1 \\ &= 6 \times 0.737 \times 0.05 \\ &= 0.2321 \end{aligned}$$

(iii) Find mean no of letters that will arrive within two days.

$$\begin{aligned} \mu &= np \\ &= 6(0.95) \\ &= 5.7 \end{aligned}$$

(iv) Compute variance and SD of the no that will arrive within two days?

$$\begin{aligned} \sigma &= npq \\ &= 6 \times 0.95 \times 0.05 \\ &= 0.285 \end{aligned}$$

$$\begin{aligned} \sigma^2 &= \sqrt{0.285} \\ &= 0.5339 \end{aligned}$$